

Plate to Platform: Swiggy vs Zomato

Comprehensive Analysis Across Consumer Tiers, Economics, Governance, and Optimization

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1. PLATFORM DEMOGRAPHICS DASHBOARD INSIGHTS

Author Insights (Observed) Foundational Mid-Range Dominance: Moderate segment (₹300–₹700) constitutes ~50% of aggregate platform revenue (~₹110.8 Cr), anchoring the entire food delivery ecosystem. Zomato's Volume Lead: Zomato establishes higher market penetration and total revenue capture in the high-frequency Moderate tier. Geospatial Clustering: Top metro order concentrations (Mumbai ~2.42L orders, Hyderabad ~1.16L, Chennai ~1.04L) drive disproportionate platform GMV.	Extended Diagnostic Insights New Discovery Cuisine Turnaround vs Rating: Continental/Italian registers longest delivery times (66 min) yet sustains high ratings (4.8), revealing customer rating resilience for premium food. 2026 Revenue Dip Anomaly: Projected revenue trend shows sharp normalization in 2026 (₹1.37 Cr from ₹2.77 Cr peak), signalling market saturation or partial data cadence. Sub-tier Distribution: Luxury (₹33.5 Cr) and Budget (₹27.1 Cr) form equal volume tails around the Moderate commercial belly.
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2. PLATFORM PERFORMANCE PER CONSUMER TIER DASHBOARD

Author Insights (Observed) Economics Segmented Profit Engines: Zomato commands net profitability across Budget Friendly (<₹300) and Moderate tiers; Swiggy outperforms in Premium (>₹700) and Elite tiers. Volume vs Realized Profit: Raw order share does not correlate with margin expansion—discount frequency severely erodes operating profit in lower tiers. Commission Realization: Platform commissions per order vary geographically (Jaipur ₹228.92, Pune ₹227.73, Kolkata ₹232.03) maintaining ~21–22% take rates.	Extended Diagnostic Insights New Discovery Merchant Discount Distortion: Bamboo Hub registers discount frequencies peaking at 62%, yet preserves a 4.6 customer rating, pointing to heavy subsidization for market share. Treemap Market Concentration: Top merchant nodes ("Bamboo Bowl", "Bamboo Bhavan", "Bamboo Bakery") capture equal 50.00 index splits, highlighting merchant stickiness. Profit Cycle Stability: Annual net profits oscillate reliably between ₹1.88 Cr and ₹2.41 Cr (2018–2025) before declining in 2026.
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Core Strategic Takeaway (Dashboards 1 & 2): Food delivery is fundamentally a bifurcated market. Zomato operates with unmatched scale and monetization in the price-sensitive and mid-market tiers, whereas Swiggy extracts its economic rent from high-AOV, premium consumer clusters.

3. SENTIMENT VS BEHAVIOUR ANALYSIS DASHBOARD

Author Insights (Observed)

Perception

- **The Sentiment–Profit Paradox:** Swiggy dominates customer sentiment across Moderate and Elite tiers, yet Zomato leads Moderate profitability and volume.
- **Budget Sentiment Inversion:** Zomato captures stronger customer sentiment in the Budget tier, driven by discount penetration and reliable delivery.
- **City Growth Divergence:** Lucknow leads monthly order growth (3.816) with 89.2 sentiment score, demonstrating Tier-2 expansion outperforming mature metros like Delhi (2.644).

Extended Diagnostic Insights

New Discovery

- **AOV Invariant Sentiment:** Platform ratings stay flat at 4.3–4.4 over the 10-year trend, whereas sentiment scores fluctuate between 85.9 and 88.2.
- **Dish Level Demand Elasticity:** Mutton Biryani (25,783 orders) and Cheesecake (25,058) represent twin high-volume pillars across divergent price points.
- **Restaurant Type Capture:** Cloud Kitchens and QSRs generate 87.3 sentiment scores with Zomato alignment, capturing highest repeat order density.

4. CONSUMER EXPERIENCE ANALYSIS DASHBOARD

Author Insights (Observed)

CX Index

- **Composite Metric Alignment:** The Min-Max normalized Consumer Experience Score (CES) affirms Swiggy's edge in Premium/Elite and Zomato's edge in Budget/Moderate.
- **Operational Correlation:** Strong linear correlation exists between delivery fee and delivery time across merchant nodes.
- **Restaurant Segment Experience:** Cloud Kitchens achieve peak CES (9.9/10), outperforming Food Trucks (8.0), QSRs (8.6), and Cafes (6.7).

Extended Diagnostic Insights

New Discovery

- **2018 Disruption Shock:** Experience score plummeted to 0.00–1.10 in 2018 before recovering to 7.31 in 2020, pointing to historic logistics or data tracking reconfigurations.
- **Dish-Specific Delight:** Green Curry (9.8) and Butter Chicken (9.1) yield industry-leading customer experience ratings, acting as high-retention anchor dishes.
- **Sentiment vs Rating Divergence:** Outlier restaurants display low ratings (1.2–1.7) despite moderate sentiment, flagging operational delivery failures.

5. FINANCIAL ANALYSIS & COMPARISON STORYBOARD

Analytical Dimension	Swiggy Ecosystem	Zomato Ecosystem	BI Strategic Conclusion
Dominant Consumer Tier	Premium & Elite (>₹700)	Budget & Moderate (<₹700)	Structural customer segmentation boundary
Cost per Order vs Commission	Linear expansion up to ₹78.2 comm.	Linear expansion up to ₹78.2 comm.	Fee, commission (~21.1%) and expenses (~23.5%) match
Financial Divergence	Superior CX & Sentiment in Mid-Tier	Superior Revenue & Net Profit in Mid-Tier	Divergence requires customer retention/CAC audit
Merchant Format Margin	Cloud Kitchen / Casual Dining	QSR & Fast Food Partnerships	Different merchant mix drives final EBITDA profile

Methodological Governance Note: Delivery fees, commissions, and expense structures show near-identical distribution across both platforms. Therefore, revenue divergence cannot be causally attributed to price gouging or commission disparities; platform habituation, app UI, and loyalty memberships (Swiggy One vs Zomato Gold) likely drive volume differences.

6. PERFORMANCE OPTIMIZATION: PARAMETER VS FILTER ARCHITECTURE

In analytical dashboards spanning 26 worksheets and high-density scatter plots (10,000+ data points), standard Tableau dimension filters trigger severe query bottlenecks. The architecture was optimized by replacing global dimensional filters with parameterized evaluation engines.

Traditional Quick Filter Deficiencies	Bottleneck	Implemented Parameter Architecture	Optimized
Sequential Database Queries: Each dimensional quick filter generates independent <code>SELECT DISTINCT</code> queries to populate domain lists. Context Filter Query Overhead: Cross-worksheet filtering invalidates Tableau's workbook cache, forcing re-aggregations across all 26 views. Render-Thread Locking: Multi-select domain trees inflate DOM element counts in Tableau Server/Public screen containers.		In-Memory Evaluated Logic: Parameters hold constant values in memory; switches evaluate single-pass Boolean calculations. Consolidated Dashboard Footprint: 3 Master Parameters (Platform Selection, Consumer Tier, Selected Metric) replaced over 15 discrete filters. Cache Retention: Parameter value changes do not re-query schema metadata, preserving Tableau VizQL data cache and boosting FPS.	

Optimized Calculated Logic Implementation

Field Name	Optimized Calculation Architecture	Performance Impact
[Selected Platform Revenue]	<pre>CASE [Platform Selection] WHEN 'SWIGGY' THEN [Swiggy Est. Revenue] WHEN 'ZOMATO' THEN [Zomato Est. Revenue] ELSE [Swiggy Est. Revenue] + [Zomato Est. Revenue] END</pre>	Eliminates table joins; computes dynamically via column projection in single scan.
[Tier Filter Flag]	<pre>[Platform Selection] = 'ALL' OR ([Consumer Tier] = 'Budget Friendly' AND [Avg Cost Per Order] < 300) OR ([Consumer Tier] = 'Moderate' AND [Avg Cost Per Order] >= 300 AND [Avg Cost Per Order] < 700)...</pre>	Boolean evaluation placed on Filter shelf set to TRUE; triggers early row-drop in execution pipeline.

7. TABLEAU PREP & DATA ENGINEERING PIPELINE

Data readiness was executed through an initial spreadsheet restructuring workflow combined with Tableau-ready normalization protocols.

Phase 1: CSV Cleansing (Excel) Manual Data Scrubbing: - Corrected corrupted delimiters, trailing spaces, and character encoding issues. - Structured column headers into normalized relational nomenclature. - Handled null values in numerical fields (cost, commission, fees).	Phase 2: Mathematical Normalization Min-Max Scaling Layer: - Mapped raw ratings & sentiment to a standard 0–10 scale. - Zero-division safeguard: if Max = Min, defaulted index to 5.0. - Synthesized composite Consumer Experience Score: $(CSI + CRI) / 2$.	Phase 3: Hyper Extract Generation High-Speed Ingestion: - Materialized calculated fields directly into Tableau Hyper format. - Pre-aggregated transaction metrics to optimize workbook responsiveness. - Configured schema for real-time parameter switches.
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Data Engineering Pipeline Summary: Raw CSV cleaning and structure formatting were established first in Excel to guarantee data integrity, followed by automated normalization and Hyper extract compilation to enable lag-free parameter querying across Tableau Public.

8. DATA GOVERNANCE & METRIC INTEGRITY FRAMEWORK

To ensure enterprise-grade reliability and regulatory compliance across dual-platform reporting, the following governance standards are established:

Single Source of Truth (SSOT) Dictionary

- **Revenue Definition:** Estimated Gross Merchandise Value (GMV) post-cancellation, excluding uncollected platform tips.
- **Commission Amount:** Standardized as $\text{Commission \%} \times \text{Avg Cost Per Order}$; blended as weighted average for combined views.
- **Expense Approximation:** Modeled derived field: $\text{Revenue} - \text{Estimated Net Profit}$ (representing delivery, tech, CAC, and ops overhead).
- **Net Profit Margin:** Uniformly evaluated as $\text{Total Net Profit} / \text{Total Revenue} \times 100$.

Quality Auditing & Data Lineage

- **Divergence Disclaimers:** Modeled metrics explicitly labeled as "Estimated" to prevent misinterpretation as audited GAAP financials.
- **Boundary Integrity:** Outlier filtering applied to delivery fees ($>₹150$ flagged) and delivery times (>120 min flagged).
- **Version Control:** Standardized calculation naming conventions (e.g., [Selected Platform ...] prefix for all parameter-dependent fields).

9. STAKEHOLDER COMMUNICATION STRATEGY

Different executive tiers require tailored analytical narratives derived from this BI system:

Executive Persona Communication Matrix

Stakeholder Tier	Primary Business Concern	Target Dashboards	Executive Narrative & Action Plan
C-Suite / Leadership (CEO, CFO, COO)	EBITDA margin, market share retention, segment profitability.	Demographics (D1), Financial Analysis (D5), Storyboard	Focus on defending the Moderate revenue engine (~50% GMV). Investigate why Swiggy's high sentiment in Mid-Range fails to convert to Zomato's profit levels.
VP of Growth & Marketing	AOV progression, discount burn, city expansion.	Performance per Tier (D2), Sentiment vs Behaviour (D3)	Reallocate promotional discounts away from saturated metros toward high-growth Tier-2 nodes (Lucknow, Hyderabad) with high sentiment responsiveness.
Head of Merchant Partnerships	Commission yield, restaurant retention, delivery SLAs.	Consumer Experience (D4), Financial Analysis (D5)	Standardize commission tiers around Cloud Kitchens (9.9 CES); support Italian/Continental merchants with dedicated express logistics to reduce 66-min transit times.

10. STRATEGIC RECOMMENDATIONS & ROADMAP

Actionable Recommendations

- **Investigate Moderate Segment Gap:** Commission a dedicated cohort analysis examining Customer Acquisition Cost (CAC) and loyalty subscription churn.
- **Rationalize Discounting:** Cap merchant discounts on venues with $>50\%$ discount frequency where rating uplift is negligible.
- **Capitalize on High-AOV Tiers:** Deploy premium packaging and priority dispatch for orders $>₹1,300$ to reinforce Swiggy's margin moat.

Next-Phase BI Enhancements

- **Tableau Prep Automated Flows:** Transition manual Excel CSV cleaning into a scheduled, containerized Tableau Prep Conductor pipeline.
- **Predictive Churn Modeling:** Integrate Python/R TabPy scripts to forecast customer lifetime value (CLV) based on CES deviations.
- **Dynamic What-If Scenarios:** Introduce parameter-driven commission sensitivity simulators for merchant contract negotiations.